

# DengAI — Predicting Disease Spread

## Metrics and Models

October 2026

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### Metrics

The competition metric is MAE

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

The average number of cases we are off by, per week.

- Reported over all **416 test weeks** (260 San Juan + 156 Iquitos)
- Lower is better; a perfect model scores 0
- Both cities are pooled into one number, so **San Juan dominates** — it has 63% of the test weeks *and* roughly four times the case counts

### Why MAE and not RMSE

The target is extremely skewed:

|                 | San Juan    | Iquitos     |
|-----------------|-------------|-------------|
| mean cases/week | 34.2        | 7.6         |
| <b>median</b>   | <b>19</b>   | <b>5</b>    |
| max             | 461         | 116         |
| <b>skewness</b> | <b>4.48</b> | <b>4.00</b> |

In both cities the **worst 10% of weeks hold 43% of all cases**.

## What “skewed” actually means

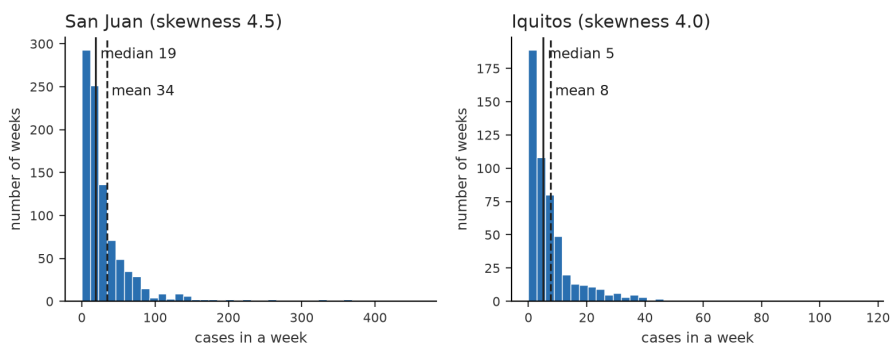


Figure 1: Weekly dengue cases: most weeks are quiet, a few outbreak weeks form a long tail

## What “skewed” actually means (cont.)

Most weeks have few cases; a handful of outbreak weeks have hundreds. The distribution is lopsided, with a **long tail to the right**.

- **The mean gets pulled toward the tail.** San Juan’s mean is 34 cases, but a typical (median) week has 19. About **73% of weeks are below the mean**.
- **Skewness is a number for this lopsidedness.** 0 means symmetric, like a bell curve; above 1 already counts as strongly skewed. Ours is **4.5 and 4.0**.
- **Everyday analogy:** income. A few billionaires raise the *average* income, but the *median* still describes the typical person.

For us this means that the rare outbreak weeks would dominate any metric that punishes big errors heavily.

## Why MAE and not RMSE (cont.)

RMSE squares the errors, so a single epidemic week with 461 cases contributes as much as **~500 ordinary weeks** being off by one.

- Under RMSE the model is dragged into fitting a handful of outbreaks
- MAE weights every week equally
- Statistically: **MAE is minimised by the conditional median, RMSE by the mean** — and for a skewed count distribution the median is the more robust target

This also explains a behaviour we see later: good models here are *conservative*.

## Models

### The pipeline has four branches

Different model families need different preprocessing, so `make_city_pipeline` builds one of four shapes:

| Branch                   | Steps                                              | Models                          |
|--------------------------|----------------------------------------------------|---------------------------------|
| <code>tree</code>        | impute → history →<br>cyclical → select →<br>model | CatBoost,<br>RandomForest       |
| <code>dense</code>       | tree steps + <b>impute</b> +<br><b>scale</b>       | Ridge, Lasso                    |
| <code>time_series</code> | impute → history →<br>date + scaled weather        | Prophet, SARIMAX                |
| <code>seasonal</code>    | calendar columns only                              | Seasonal median,<br>Shape×level |

### Why `dense` exists

The `tree` branch leaves **NaNs** in the lag/rolling warm-up — the first 26 weeks have no 26-week history.

- Tree models (sklearn  $\geq 1.4$ ) **handle NaN natively**
- RidgeCV raises `ValueError: Input X contains NaN`

So linear models get two extra steps: `SimpleImputer(median)` then `StandardScaler`. Scaling matters for them too, since Ridge and Lasso penalise coefficients and are not scale-invariant.

### Why `seasonal` exists

The tree branch **drops** `city`, `week_start_date` and `weekofyear` before the model.

But the two calendar baselines need exactly those columns. Rather than weaken the main branch, they get their own: a `ColumnTransformer` passing through `year` and `weekofyear`, nothing else.

These models **never see the weather at all** — which is the point: they are the baseline any weather-driven model should have to beat.

### The eight models

| Model    | Family            | Core assumption                              |
|----------|-------------------|----------------------------------------------|
| CatBoost | gradient boosting | non-linear interactions of<br>lagged weather |

| Model                | Family               | Core assumption                                            |
|----------------------|----------------------|------------------------------------------------------------|
| RandomForest         | bagged trees         | same, without boosting                                     |
| Ridge                | linear + L2          | smooth linear response, all features shrunk                |
| Lasso                | linear + L1          | linear, but most features are irrelevant                   |
| Prophet              | additive time series | trend + annual seasonality + regressors                    |
| SARIMAX              | ARIMA + exog         | autocorrelated errors + Fourier seasonality                |
| Seasonal median      | calendar only        | this week behaves like this week in past years             |
| Shape $\times$ level | calendar only        | seasonal <i>shape</i> $\times$ typical annual <i>total</i> |

## Results — holdout MAE

| Model                                  | San Juan     | Iquitos     |
|----------------------------------------|--------------|-------------|
| <b>CatBoost</b>                        | <b>15.02</b> | 6.27        |
| RandomForest                           | 15.16        | 8.39        |
| <b>Shape <math>\times</math> level</b> | 20.72        | <b>3.39</b> |
| Lasso                                  | 21.16        | 4.50        |
| Prophet                                | 21.78        | 7.16        |
| Seasonal median                        | 23.49        | 4.41        |
| Ridge                                  | 24.46        | 5.48        |
| SARIMAX                                | 24.56        | 4.61        |

Different winners per city — so the final submission uses **CatBoost for San Juan**, **SARIMAX for Iquitos**.

## Side experiment — XGBoost and feature selection

A separate notebook (`experiments/xgboost_feature_selection.ipynb`) asks whether all 127 engineered features help, or whether fewer would do: XGBoost tuned on chronological CV (3 folds of 52 weeks) with three feature-selection strategies.

| Feature selection           | San Juan                  | Iquitos                  |
|-----------------------------|---------------------------|--------------------------|
| All 127 features            | 17.8                      | 7.2                      |
| XGBoost importance, top 100 | 17.0                      | 7.1                      |
| <b>Ridge RFE</b>            | <b>14.5</b> (50 features) | <b>6.7</b> (10 features) |

CV MAE. Recursive elimination with a *linear* model chose features better than the trees' own importances did.

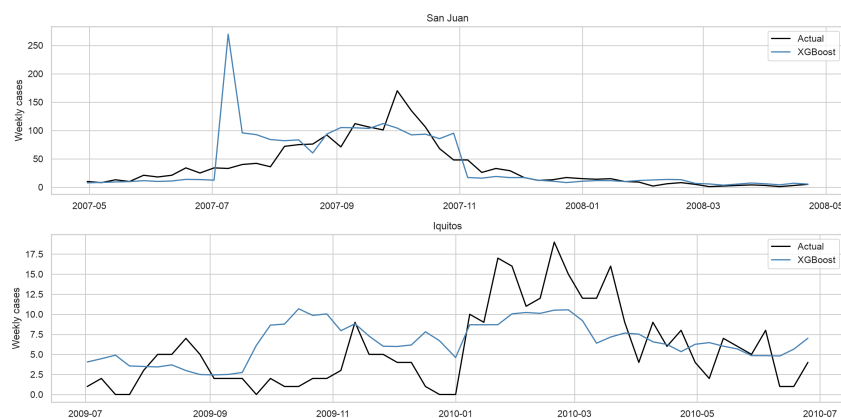


Figure 2: XGBoost (blue) against actual cases (black), 52-week holdout

## XGBoost on the holdout year

Holdout MAE **17.21** in San Juan (third; CatBoost 15.02) and **3.54** in Iquitos: the **best weather-based model** there (SARIMAX 4.61, CatBoost 6.27), with only ten long-window features. It follows the 2007 outbreak but under-predicts both peaks.

## What drives the CatBoost predictions?

Share of CatBoost feature importance, grouped by what the feature measures

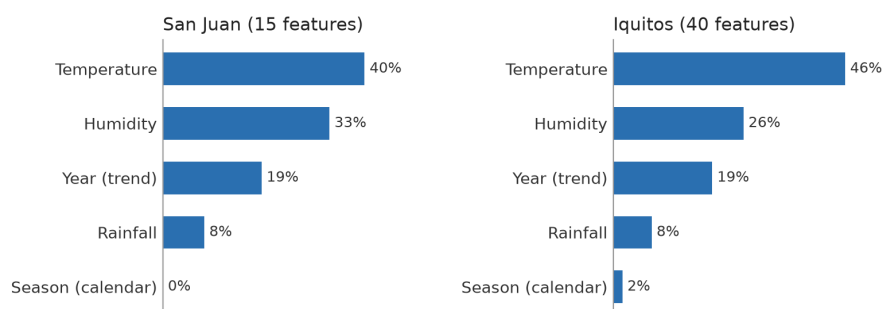


Figure 3: CatBoost feature importance, grouped by what each feature measures

## Feature importance — what the model is telling us

- **Temperature and humidity carry about 70% of the importance** in both cities. Mosquitoes breed and bite more when it is warm and humid.
- **Every top feature is a long average over 8–26 weeks**, not this week's

weather. Weather acts with a delay: the mosquito population has to build up first.

- **Rainfall matters less than expected** (about 8%). Temperature and humidity already capture most of what the rain does.
- **year alone takes about 19%**. Case levels drift from year to year for reasons that are not weather.

Importance tells us what the model *uses*, not what *causes* dengue.

## Why weather is not the whole story

Feature importance and the results table point the same way. In Iquitos the calendar-only **Shape × level** model (3.39) beats every weather model:

- Climate reliably predicts **when** cases rise within a year
- It explains very little about **how many**

Epidemic magnitude is driven by which dengue **serotype** is circulating and by population immunity — neither is in this dataset.

So a model that predicts the *shape* from the calendar and refuses to guess the *level* from weather is not naive. It is **honest about what the data supports**.

## Summary

**Metrics** — MAE, because the target is skewed (skewness 4.5): a few outbreak weeks would dominate RMSE.

**Models** — eight models across four pipeline branches, because tree, linear, time-series and calendar families need different preprocessing. Winners differ by city. A side experiment with XGBoost confirms that feature selection matters: 50 features beat 127 in San Juan, and 10 are enough in Iquitos.

**Feature importance** — CatBoost relies mostly on temperature and humidity averaged over months. Weather tells us *when* cases rise, not *how many*.